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Fourth Year Engineering Project
Preliminary Design

Classification of objects at a very low resolution

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10.4.2.3 Abstract

The project focuses on examining the effect of decreased image quality on the accuracy of object classification methods, based on both deep learning and classical approaches. The need for this project arises from the difficulty of identifying objects in low-quality images, particularly under conditions that simulate Terahertz (THz) imaging, where image quality is naturally limited by low resolution and high noise. The aim of this work is to evaluate the extent to which classification accuracy can be maintained under challenging conditions by analyzing the robustness of existing models to different types of quality degradation. Our primary objective is to achieve an average accuracy of approximately 80% on degraded data while finding the most effective network structures for information preservation. The innovation of this approach focuses on leveraging biomimetic "Aggregated Attention" and "Feature Reuse" as advanced solutions for dealing with data loss in situations of limited visual information. The methodology is based on using well-known datasets, including MNIST and CIFAR-10, and generating low-quality versions through blurring, noise addition, and color removal to compare a ResNet-50 baseline against DenseNet and TransNeXt architectures . It is expected that through these advanced features, performance can be significantly improved compared to baseline methods, proving that smart model design can overcome severe image quality limitations.

Keywords: object classification, deep learning, neural networks, low resolution, noise, blur, grayscale, THz imaging, robustness, feature reuse.

תקציר

הפרויקט עוסק בבחינת השפעת ירידה באיכות התמונה על רמת הדיוק של שיטות סיווג עצמים, המבוססות על למידה עמוקה ושיטות קלאסיות. הצורך בפרויקט נובע מהקושי לזהות עצמים בתמונות באיכות ירודה, ובפרט במצבים המדמים תנאים של תמונות מתדרי טרה-הרץ (THz), שבהן איכות התמונה נמוכה באופן טבעי בשל רזולוציה מוגבלת ורעש רב. מטרת העבודה היא לבחון עד כמה ניתן לשמור על דיוק הסיווג גם בתנאים מאתגרים, תוך ניתוח עמידותן של שיטות קיימות לפגיעות שונות באיכות התמונה. היעד המרכזי של המחקר הוא השגת רמת דיוק ממוצעת של כ-80% על נתונים משובשים תוך מציאת מבני הרשת היעילים ביותר לשימור מידע. החדשנות בפרויקט מתמקדת בשימוש בטכניקות של "קשב ביומימטי" ו"שימוש חוזר בתכונות"- כפתרונות מתקדמים להתמודדות עם אובדן נתונים במצבים של חוסר במידע חזותי. הפרויקט מבוסס על שימוש במאגרי נתונים מוכרים, בהם MNIST ו-CIFAR-10, ועל יצירת גרסאות באיכות ירודה הכוללות טשטוש, הוספת רעש והורדת צבע לצורך השוואה מבוקרת בין מודל בסיס ResNet-50 לבין הארכיטקטורות DenseNet ו-TransNeXt . הציפייה היא כי גם לאחר ירידה באיכות התמונה, ניתן יהיה לשפר את ביצועי המודלים ביחס לשיטות הבסיס ולהוכיח כי תכנון חכם של ארכיטקטורת הרשת יכול לפצות על מגבלות חמורות באיכות התמונה.

מילות מפתח: סיווג עצמים, למידה עמוקה, רשתות נוירונים, רזולוציה נמוכה, רעש, טשטוש, שחור-לבן, טרה-הרץ, חסינות מודל, שימוש חוזר בתכונות.

10.4.2.4 Project Objectives

The primary goal of this project is to achieve high classification accuracy for objects in images with very low resolution and significant noise, simulating the output of Terahertz (THz) imaging sensors.

List of Goals and Success Metrics:

1. **Achieving High Accuracy on Degraded Data (Main Goal):**
    - **Objective:** To successfully classify objects in low-quality inputs (blurred, noisy, grayscale) where standard visual details are missing.
    - **Success Metric:** Achieve a Top-1 accuracy of at least 80% on the "Simulated THz" benchmark (degraded CIFAR-10), demonstrating that the system is viable for real-world applications.

2. **Minimizing Performance Drop (Robustness):**
    - **Objective:** To create a system that maintains its performance stability even when image quality deteriorates significantly compared to the original source.
    - **Success Metric:** The accuracy drop between the original high-quality images and the low-quality inputs should be less than 15%.

3. **Identifying the Optimal Architecture:**
    - **Objective:** To find the specific neural network architecture that best solves this problem. Following a research phase that included candidates like TResNet and EfficientNetV2, we have selected TransNeXt as the primary challenger to DenseNet. This allows us to determine which mechanism (Feature Reuse (DenseNet) or Aggregated Attention (TransNeXt)) yields the highest success rate.
    - **Success Metric:** Identification of a specific model that outperforms the baseline (ResNet50) by a statistically significant margin (e.g., >5%).

4. **Feasibility for Edge Implementation:**
    - **Objective:** To ensure the chosen solution is efficient enough for potential deployment in sensor systems.
    - **Success Metric:** The final selected model should achieve an inference time of less than 50ms per image on a standard GPU.

10.4.2.6 Research Proposal for a Research Project

Project Name:
Classification of Objects at a Very Low Resolution

Problem Description:
Standard object classification systems fail when applied to low-quality imaging sensors, such as Terahertz (THz) cameras. These sensors produce images with low spatial resolution, high noise, and no color. Because standard Convolutional Neural Networks (CNNs) rely on high-frequency details (textures and sharp edges) that are lost in these conditions, their classification accuracy drops drastically. The problem this project aims to solve is the inability of current systems to reliably identify objects when visual information is severely compromised.

Methodology and Technology:
The research will be conducted using CIFAR-10 and MNIST datasets with synthetic degradation (Gaussian blur, noise, grayscale). We propose a hierarchical investigation based on the hypothesis that preventing information loss is more critical than network depth in this domain. Our investigation will test three tiers of architectural evolution:

1. **Baseline (Standard CNN):** We will use ResNet50 to establish the performance floor. We expect this model to degrade significantly due to its reliance on texture and standard pooling layers which discard spatial information.
2. **Intermediate Solution (Feature Reuse):** We will implement DenseNet. We hypothesize that its Feature Reuse mechanism will allow it to outperform ResNet by preserving low-level features from early layers, which are less affected by the degradation.
3. **Optimal Solutions (SOTA Candidates):** We have reviewed three state-of-the-art architectures: TResNet, EfficientNetV2, and TransNeXt, which utilize advanced mechanisms to handle degraded data.
    - **TResNet:** Uses a SpaceToDepth stem to preserve input information.
    - **EfficientNetV2:** Uses Progressive Learning for scale robustness.

- **TransNeXt:** Uses Biomimetic Aggregated Attention to focus on structural details.

Implementation Strategy: To ensure project feasibility within the academic timeline and available GPU resources, we will focus our primary implementation and training on TransNeXt as the advanced challenger. This selection allows us to directly compare the effectiveness of Biomimetic Attention against the Feature Reuse approach of DenseNet.

Mathematical Model:

The classification task is modeled as finding the optimal parameters θ that minimizes the prediction error. Let x be the clean image and y be the true class label. The degraded image $\bar{x}$ is defined as:

$$\bar{x} = D(x; \phi) + n$$

Where $D(\cdot)$ is the degradation operator (blur/down-sample) parameterized by severity ϕ, and n is additive noise. The classifier $f(\cdot; \theta)$ learns to predict y by minimizing the Cross-Entropy Loss function L:

$$\theta^* = argmin_\theta \sum L(f(\bar{x}; \theta), y)$$

$$L(\hat{y}, y) = -\sum_c y_c log(\hat{y}_c)$$

Research Stages:

1. **Data Simulation:** Building the pipeline to convert standard datasets into "Low Resolution/THz" datasets with controlled severity levels.
2. **Baseline Testing:** Running standard ResNet50 to measure the exact failure point (baseline accuracy).
3. **Optimization Phase:** Implementing the proposed architectures (DenseNet and the selected challenger, TransNeXt) and tuning them specifically for the low-quality dataset.
4. **Selection:** Selecting the single best-performing model based on the highest accuracy achieved.
5. **Final Validation:** Testing the winning model on a separate test set to confirm it meets the 80% accuracy goal.

Expected Results:

We expect to demonstrate a clear performance hierarchy where the SOTA models (TResNet, TransNeXt, EfficientNetV2) outperform DenseNet, which in turn outperforms the ResNet baseline. Specifically, we anticipate that the Biomimetic Aggregated Attention mechanism of TransNeXt will provide the most significant boost in robustness by effectively modeling local pixel neighborhoods and structural details.

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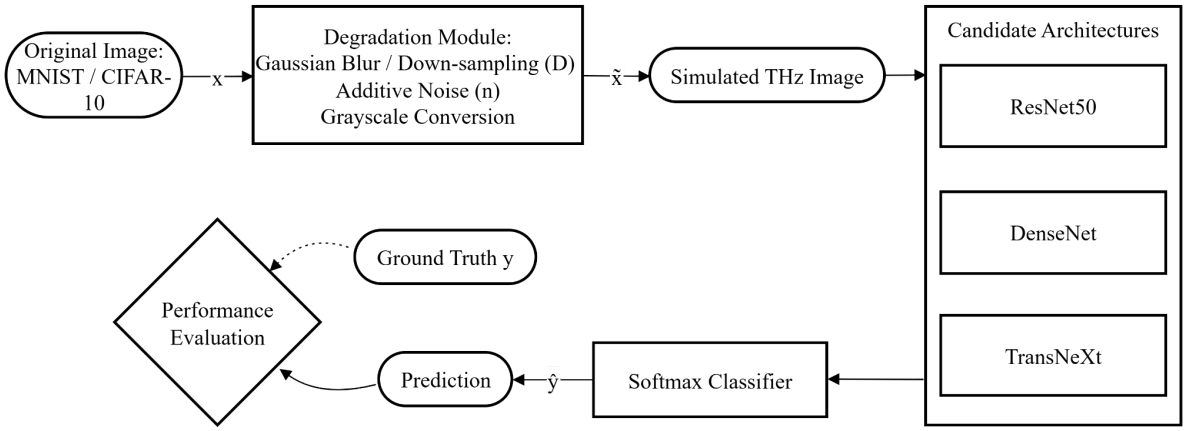
Potential Applications:

- **Security:** Threat detection in low-res body scanners.
- **Medical:** Diagnostics in noisy ultrasound/microscopy.
- **Automotive:** Object detection in heavy fog or rain.

Block Diagram Description:



10.4.2.7 Literature Review

Introduction and Basic Concepts:

The field of image classification has undergone a revolution with the advent of Deep Learning and specifically Convolutional Neural Networks (CNNs). A standard CNN architecture typically consists of a series of convolutional layers, non-linear activation functions (such as ReLU), and pooling layers. These layers work hierarchically to extract features from the input image: initial layers detect simple patterns like edges and textures, while deeper layers combine these into complex shapes and semantic objects.

While standard CNNs achieve high accuracy on high-quality datasets like ImageNet, the problem addressed in this project- classification of objects at very low resolution and high noise levels (simulating Terahertz imaging)- poses unique challenges. In such degraded conditions, the fine details required for high-level feature extraction are often lost or obscured. Therefore, the choice of network architecture is critical, specifically its ability to preserve and reuse information throughout the network depth.

Classical and Standard Approaches:

Early deep learning models established the foundations for modern architectures but have limitations regarding the specific constraints of this project.

- **AlexNet**: This model marked a pivotal moment in computer vision by winning the ILSVRC-2012 competition. It utilizes five convolutional layers and introduced the use of ReLU to solve the vanishing gradient problem. However, its relatively shallow architecture limits its capacity to extract meaningful representations from highly corrupted or low-information images.

- **VGG:** The VGG architecture increased the depth of the network (16-19 layers) and standardized the use of small 3×3 convolutional filters. While robust, VGG is computationally expensive and prone to overfitting on smaller datasets without extensive regularization. Furthermore, as a plain feed-forward network, it suffers from information loss as signals pass through many layers without shortcuts.

- **ResNet (Residual Networks):** ResNet introduced the concept of "skip connections" (or identity shortcuts), allowing the training of extremely deep networks (hundreds of layers) by mitigating the vanishing gradient problem. In a Residual Block, the input x is added to the output of the convolutional layers $F(x), i.e., y = F(x) + x$.

- **Analysis for Low-Resolution:** While ResNet allows for deep feature hierarchies, the summation operation (element-wise addition) can potentially impede information flow. By collapsing the features from previous layers with the current layer's output, subtle "low-level" features- which are crucial when the image resolution is too low to form distinct high-level shapes- might be "washed out" or lost.

Advanced Architectures for Feature Preservation:
To address the limitations of information loss in deep networks, alternative connectivity patterns have been proposed.

- **DenseNet (Densely Connected Convolutional Networks):** DenseNet proposes a different connectivity pattern: instead of summing features like ResNet, it concatenates them. In a Dense Block, each layer receives the feature maps of all preceding layers as input.
    - **Mechanism:** The lth layer receives inputs $\left[x_0, x_1, ..., x_{l-1}\right]$, ensuring that the "collective knowledge" of the network is explicitly preserved.
    - **Relevance:** This property is theoretically ideal for low-resolution classification. When high-level semantic features cannot be reliably formed due to blur or noise, the network can still rely on the preserved low-level textures and edges to make a prediction. Furthermore, DenseNet is highly parameter-efficient compared to ResNet.

- **NASNet (Neural Architecture Search):** Instead of relying on manual architecture design, NASNet utilizes a search algorithm (based on reinforcement learning) to discover the optimal structure of a convolutional "cell" on a smaller dataset (CIFAR-10), which is then stacked to form the full network.
    - **Relevance:** The search space allows for diverse operations (e.g., separable convolutions, pooling). The resulting architectures often outperform human-designed models and may discover non-intuitive combinations of operations that are inherently more robust to noise or small spatial dimensions.

Modern Efficiency-Focused Architectures:

Recent advancements focus on optimizing the trade-off between speed, accuracy, and robustness, often introducing techniques relevant to handling low-quality data.

- **EfficientNetV2:** EfficientNetV2 improves upon the original EfficientNet by optimizing training speed and parameter efficiency. A key innovation presented is Progressive Learning.
    - **Mechanism:** During training, the image size and regularization strength (e.g., data augmentation) are dynamically adjusted. The model starts with smaller images and weak regularization and progressively moves to larger images with stronger regularization.
    - **Relevance:** This approach is highly relevant to the project's goal. By exposing the model to varying resolutions and regularization intensities during training, the network learns to be robust to scale changes and information loss.

- **TResNet:** TResNet focuses on maximizing GPU throughput while maintaining high accuracy. It introduces several architectural refinements, including a SpaceToDepth stem.
    - **Mechanism:** Instead of using aggressive pooling layers at the beginning of the network (which discard spatial information), TResNet uses a SpaceToDepth layer that rearranges spatial blocks into the depth dimension, preserving all input information.
    - **Relevance:** In low-resolution images, every pixel contains vital information. Standard pooling layers might discard the few remaining distinctive features. TResNet's approach attempts to minimize this information loss, potentially offering better performance on small, blurry inputs.

Vision Transformers and Biomimetic Designs:

The most recent wave of architectures involves Vision Transformers (ViTs). While standard ViTs are computationally heavy, newer variants offer compelling properties for robust perception.

- **TransNeXt:** TransNeXt introduces a biomimetic design termed "Aggregated Attention" which simulates biological foveal vision.

- **Mechanism:** It employs "Pixel-focused Attention", where each query has fine-grained attention to its nearest neighbors (simulating foveal vision) and coarse-grained attention to global features (simulating peripheral vision).
- **Relevance:** TransNeXt achieves state-of-the-art results on robustness benchmarks (ImageNet-A). The explicit modeling of local vs. global context is crucial when the global shape is distorted by low resolution. The ability to focus intensely on local pixel neighborhoods might allow the extraction of features that are otherwise lost in standard convolution or global self-attention layers.

Summary:

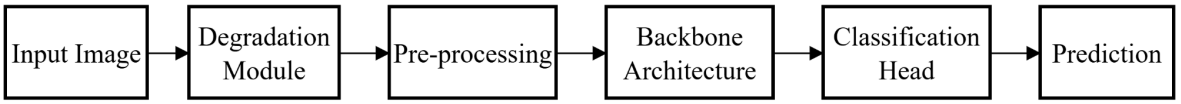
The literature survey highlights a progression from standard CNNs to architectures designed for specific challenges like feature preservation and robustness. While ResNet serves as a strong standard baseline, its summation mechanism poses a risk of feature "washout" in low-information scenarios. DenseNet addresses this theoretically via feature reuse, making it a primary candidate for this project. However, modern architectures like TransNeXt (with robust foveal attention), TResNet (with information-preserving downsampling), and EfficientNetV2 (with progressive learning for scale robustness) offer alternative sophisticated mechanisms to handle data degradation. The proposed work will therefore focus on evaluating DenseNet against these baselines to verify the hypothesis regarding feature preservation in low-resolution regimes.

10.4.2.8 Design Proposal

The proposed solution addresses the challenge of classifying objects in images with severe quality degradation (low resolution and high noise) by leveraging deep learning architectures designed for maximum information preservation. The primary hypothesis is that in such degraded conditions, high-frequency details and complex semantic features are often lost early in standard feed-forward networks. Therefore, the design focuses on DenseNet (Densely Connected Convolutional Networks), which facilitates "feature reuse" by connecting every layer to all preceding layers. This ensures that the classifier retains access to raw, low-level features (such as edges and textures) that might be the only reliable information remaining in the noisy input. Additionally, advanced attention-based models like TransNeXt will be implemented to evaluate the efficacy of biomimetic foveal perception in robustly handling noise.

System Block Diagram:

The system is designed as a sequential processing pipeline. Below is the description of the schematic blocks:



Block Descriptions:

- **Degradation Module:** This component simulates the target environment (e.g., Terahertz sensor output). It accepts high-quality images from standard datasets (CIFAR-10/MNIST) and applies configurable degradation functions: down-sampling (to simulate low resolution), Gaussian blur, and additive Gaussian/Speckle noise to reduce the Signal-to-Noise Ratio (SNR).

- **Pre-processing:** Prepares the degraded data for the network. Operations include resizing (up-sampling if necessary to match network input requirements) and normalization (adjusting pixel intensity distribution).

- **Backbone Architecture (Feature Extractor):** The core algorithmic component is designed as a modular block, allowing for the systematic comparison of different architectures. We detail the DenseNet implementation below as a representative

architecture. However, to fulfill the comparative goals of this research, this backbone is designed to be swapped with potential candidates such as TResNet, TransNeXt, and EfficientNetV2, for the scope of this research and considering academic GPU constraints, the implementation will focus on evaluating DenseNet against TransNeXt.

- **Classification Head:** Converts the extracted feature maps into class probabilities. This typically consists of a Global Average Pooling layer followed by a Fully Connected (Linear) layer and a Softmax activation function.
- **Prediction:** Outputs the predicted class label with the highest probability.

Algorithmic Description:

Since DenseNet serves as our primary candidate for evaluating feature preservation, we provide its mathematical formulation here. Other candidates (TResNet, TransNeXt, EfficientNetV2) will replace this internal logic while maintaining the same input/output interfaces. Unlike traditional networks where the lth layer connects only to the $(l + 1)^{th}$ layer, or ResNet where features are combined via summation ($y = F(x) + x$), DenseNet employs concatenation.

Mathematical Model:

For each layer l, the input x_l is the concatenation of the feature maps from all preceding layers $0, \ldots, l - 1$:

$$x_l = H_l([x_0, x_1, \ldots, x_{l-1}])$$

where $[x_0, \ldots, x_{l-1}]$ represents the concatenation operation.

Key Components:

- **Composite Function (H_l):** Defined as a sequence of three operations: Batch Normalization (BN), Rectified Linear Unit (ReLU), and a 3×3 Convolution. This order (Pre-activation) helps with training deep networks.
- **Growth Rate (k):** A hyperparameter regulating the amount of new information each layer contributes to the "global state" of the network. A relatively small growth rate (e.g., $k = 12$ *or* $k = 32$) is sufficient, making the model parameter-efficient.

- **Transition Layers:** Since feature map sizes must remain constant for concatenation, the network is divided into dense blocks. Between blocks, transition layers perform 1×1 convolution and 2×2 average pooling to down-sample the spatial dimensions.

Alternative Approach (TransNeXt):
To challenge the CNN baseline, the design includes TransNeXt, which utilizes Pixel-focused Attention. This algorithm calculates attention weights based on nearest neighbors (simulating foveal vision) and global pooling (simulating peripheral vision). This dual-path mechanism explicitly models local texture robustness, which is critical when global shape information is corrupted.

Project Constraints:

- **Computational Resources:** Training DenseNet and Transformer-based models is memory-intensive due to the concatenation of feature maps and attention mechanisms. The project is limited to available academic GPU resources (e.g., Colab Pro or Departmental servers), which restricts the maximum batch size and model depth.

- **Data Availability:** There is no large-scale, labeled dataset of real-world Terahertz images available for training from scratch. Therefore, the project is constrained to using Simulated Degradation on standard datasets (CIFAR-10, MNIST) as a proxy for real sensor data.

- **Time:** The project must be completed within two academic semesters, limiting the scope of hyperparameter tuning and the number of distinct architectures that can be fully implemented and trained to convergence.

Project Assumptions:

- **Simulation Validity:** It is assumed that the mathematical degradation models (Gaussian blur, down-sampling, AWGN) sufficiently approximate the physical challenges of low-quality sensors (like THz imaging). If the real-world physics involves complex non-linear distortions not modeled here, the results may not fully generalize.

- **Feature Hierarchy:** The design assumes that preserving low-level features (edges/textures) is the key to classifying low-resolution images. If the classification task relies heavily on high-level semantic context that is destroyed by the resolution drop, preserving low-level features alone may not yield the expected 80% accuracy.

Initial Risks:

1. **Overfitting to Noise:** Given the low information content in the input images, deep networks may learn to classify based on the specific noise patterns of the training set rather than the object features.
    - Mitigation: Implementation of strong regularization techniques (Dropout, Weight Decay) and aggressive data augmentation (e.g., Cutout, Mixup) as suggested in modern training recipes.
2. **Training Latency:** DenseNet's high memory access cost can lead to slow training times compared to ResNet.
    - Mitigation: Utilizing "Memory-Efficient" implementations of DenseNet or exploring efficient variants like TResNet or EfficientNetV2 for comparison if training times become prohibitive.
3. **Performance Ceiling:** There is a risk that the information loss in the "Simulated THz" images (extreme down-sampling) is theoretically irreversible, making high-accuracy classification impossible regardless of the architecture.
    - Mitigation: Conducting a preliminary "human performance" test on the degraded images to establish a realistic upper bound for accuracy, and adjusting the degradation levels if the task proves impossible.

10.4.2.9 Definition of Project Scope

This section defines the specific boundaries of the project, detailing the final deliverables that will be handed over at the end of the development period, as well as the limitations and exclusions.

Project Deliverables:

The project aims to provide a comprehensive software solution and research insights for low-resolution image classification. The specific deliverables are:

1. **Software Simulation Package:** A complete Python-based codebase (utilizing PyTorch framework) containing:
    - **Degradation Module:** A configurable software tool to generate synthetic "low-quality" datasets from standard high-quality images (implementing down-sampling, Gaussian blur, and noise injection).
    - **Classification Models:** Fully implemented and trainable architectures for the proposed solution (DenseNet), the baseline (ResNet), and the challenger model (TransNeXt).
    - **Training & Evaluation Scripts:** Automated scripts to train the models and generate performance metrics (Accuracy, Confusion Matrix, Robustness Curves).
2. **Trained Model Weights:** The final "frozen" weights of the best-performing models, ready for inference on unseen data.
3. **Comparative Analysis Report:** A detailed document presenting the research findings, specifically:
    - Robustness curves comparing Feature Reuse (DenseNet) vs. Residual Summation (ResNet) vs. Attention (TransNeXt).
    - Recommendations for optimal architecture choice based on specific noise/resolution constraints.

Project Interfaces:

The system interfaces with the external world purely through data files:

- **Input:** Standard image files (JPEG/PNG) or Dataset objects (CIFAR-10/MNIST).
- **Output:** Classification labels (Text/Index) and performance logs (CSV/Graphs).

- **Hardware Interface:** The software runs on standard PC/Server hardware with GPU acceleration (CUDA). No specialized hardware interface is developed.

Project Boundaries (In-Scope vs. Out-of-Scope):
To ensure project feasibility within the given timeframe, clear boundaries are set:

In scope:

- Development and training of Deep Learning models.
- Simulation of sensor artifacts (noise/blur) using mathematical models.
- Evaluation of classification accuracy and inference latency on standard GPUs.
- Use of public datasets (CIFAR-10, MNIST) as proxies for real-world targets.

Out of scope:

- **Hardware Development:** The project does not involve building a physical camera, lens, or Terahertz sensor.
- **Real Data Collection:** The project will not involve collecting new raw data from physical sensors. All "low-quality" data will be synthetically generated from existing high-quality datasets.
- **Embedded Implementation:** While latency is measured, porting the code to run on specific embedded hardware (e.g., FPGA or DSP) or real-time edge devices is not part of this project's deliverables.
- **Object Detection/Segmentation:** The scope is strictly limited to Image Classification (assigning a label to an image). Tasks like bounding box detection or pixel-level segmentation are excluded.

10.4.2.10 Proposal for a Final Testing Framework

To ensure the developed classification models meet the project objectives and perform reliably under the simulated conditions of low resolution and high noise, a comprehensive testing framework has been defined. This framework outlines the testing environment, metrics, and acceptance criteria.

Product Description for Testing:
The "product" under test is a robust Deep Learning algorithm designed for image classification. Unlike standard classifiers, this model must maintain high performance even when the input image quality is severely degraded, simulating the output of a low-quality sensor (e.g., Terahertz imaging). The system takes a degraded image as input and outputs a class label.

Test Environment and Data:
The testing environment will rely on software simulations using standard datasets modified to reflect the target domain challenges.

- **Datasets:** The primary testing will be conducted on the CIFAR-10 and MNIST datasets. A held-out Test Set (10,000 images for CIFAR-10) will be strictly reserved for the final evaluation and will not be used during the training or hyperparameter tuning phases.
- **Degradation Pipeline:** To verify the model's robustness, the test set will undergo a controlled degradation process involving:
    - **Resolution Reduction:** Down-sampling images to extreme resolutions (e.g., 16x16, 8x8 pixels) and then up-sampling them back to the network's input size to simulate information loss.
    - **Noise Injection:** Adding Gaussian and Speckle noise at varying Signal-to-Noise Ratios (SNR).
    - **Blurring:** Applying Gaussian Blur with varying kernel sizes to simulate sensor defocus or diffraction limits.

Evaluation Metrics:

The performance of the models will be evaluated using the following quantitative metrics:

1. **Top-1 Accuracy:** The primary metric, measuring the percentage of images correctly classified by the model.
2. **Robustness Curve:** A plot of Accuracy vs. Noise Level (or Resolution). This visualizes how quickly the model's performance degrades as the image quality drops. A "flatter" curve indicates better robustness.
3. **Confusion Matrix:** To analyze specific inter-class errors (e.g., if the model consistently confuses "cat" with "dog" under blurry conditions).
4. **Inference Latency:** The time required to process a single image, measured in milliseconds on a standard GPU. This is crucial for potential real-time applications.

Acceptance Criteria:

The project will be considered successful if the developed models meet the following criteria:

- **Baseline Improvement:** The proposed architecture (DenseNet/TransNeXt) must achieve a higher Top-1 Accuracy than the baseline model (standard ResNet-50) on the degraded test set.
- **Absolute Accuracy Threshold:** The model should maintain an average accuracy of at least 80% on the moderately degraded test set (specific noise and blur parameters to be calibrated during the initial research phase).
- **Efficiency:** The model must converge and provide predictions within reasonable time limits on available hardware (e.g., inference time under 50ms per image).

Verification Method:

To verify the research results, a comparative analysis will be performed. The same degradation pipeline will be applied to three distinct architectures:

1. **Baseline:** ResNet-50 (Standard Industry Benchmark).
2. **Proposed Solution:** DenseNet (Feature Reuse approach).
3. **Challenger:** TransNeXt (Biomimetic Attention approach).

All models will be trained on the same training data and evaluated on the exact same degraded test instances to ensure a fair comparison. The results will be tabulated and graphed to prove which architectural approach offers superior robustness to information loss.

10.4.2.11 Budget Estimation

The following budget estimation provides a comprehensive financial plan for the project execution. As this is a research-oriented software project, the budget assumes a "Standard Industrial Model" where costs are categorized into Human Resources (engineering time) and Direct Operational Costs (computing resources and storage).
The estimation follows the guidelines for R&D tender proposals, utilizing standard market rates for junior engineers and cloud computing services.

1. Human Resources (Estimated Value):
This table estimates the value of the hours invested by the project team. The rates reflect the market cost for a Junior Algorithm Engineer (Student) and a Senior Consultant (Supervisor).

Role	Quantity	Estimated Hours	Hourly Rate (NIS)	Total Value (NIS)
Student Engineers	2	600 (300 x 2)	60	36,000
Supervisors	2	30	250	7,500
Total Human Resources				**43,500**

2. Components and Consumables:
Budget for data management and storage hardware required to handle large datasets (CIFAR-10/MNIST) and model checkpoints.

Item	Description	Quantity	Unit Cost (NIS)	Total Cost (NIS)
Storage Media	Portable SSD / Cloud Storage Subscription	1	300	300
Office Supplies	Printing, binding of reports, stationery	1	200	200
Total Components				**500**

3. Special Equipment and Computing Resources:

Costs associated with high-performance computing required for Deep Learning training (GPU acceleration).

Item	Duration	Quantity	Monthly Cost (NIS)	Total Cost (NIS)
Cloud GPU Computing	Google Colab Pro+ / AWS Instances	8 Months	~50	400
Total Special Equipment				**400**

4. Total Budget Summary:

This table summarizes the total estimated cost, including the mandatory overhead and contingency funds for unforeseen expenses.

Budget Category	Cost (NIS)
Human Resources (Theoretical)	43,500
Components & Consumables	500
Special Equipment (Computing)	400
Subtotal	**44,400**
Overhead (25%)	11,100
Miscellaneous / Unforeseen (10%)	4,440
Grand Total (Estimated Project Value)	**59,940**

10.4.2.12 Schedule and Task Allocation

This section details the project management plan, including the division of responsibilities between team members and a timeline designed to accommodate the academic calendar. The work is divided to ensure both students gain experience with the core Deep Learning components while maintaining clear individual responsibilities.

Task Allocation:

Student Name	Key Responsibilities
Bahat Itamar	1. Development of the Data Degradation Pipeline (Noise/Blur simulation). 2. Implementation and optimization of the Baseline (ResNet) architecture. 3. Technical Logistics and resource coordination.
Dotan Ann Hanna	1. Implementation of the DenseNet Architecture. 2. Development of the Evaluation Framework (metrics and graphs). 3. Documentation quality and report synchronization.
Joint Tasks	1. Training and tuning the models. 2. Implementation of the TransNeXt Challenger. 3. Analysis of final results and comparison. 4. Writing the Final Report and preparing the presentation.

Gantt Chart:

Task Name	Jan	Feb	Mar	Apr	May	Jun	Jul
Data Simulation & Setup	✓	✓					
THz-like Degradation Module	✓	✓					
Evaluation Environment Setup	✓						
Model Implementation		✓	✓	✓			
ResNet50 Baseline Training		✓	✓				
DenseNet Implementation			✓	✓			
☑ Progress Submission (15/3)			☑				
Advanced Research				✓	✓		
TransNeXt Tuning & Training				✓	✓		
☑ Poster & Abstracts (31/5)					☑		
Finalization & Submission						✓	✓
☑ Final Presentation (21/6)						☑	
Results Analysis & Final Report						✓	✓
☑ Final Submission (26/7)							☑

Detailed Timeline and Milestones:

The project lifecycle is structured into four overlapping phases, synchronized with the 2025-2026 academic milestones:

Phase 1: Data Simulation & Setup (January - February):

- Focus: Establishing the research environment and data degradation infrastructure.
- Tasks: Developing the THz-like Degradation Module (Jan-Feb) and configuring the Evaluation Environment (Jan).
- Constraint: Lower work intensity during February due to the university examination period.

Phase 2: Model Implementation (February - April):

- Focus: Deployment of core architectures and baseline verification.
- Tasks: Training the ResNet50 Baseline (Feb-Mar) followed by the implementation of the DenseNet architecture (Mar-Apr).
- Milestone: Progress Submission on March 15, 2026.

Phase 3: Advanced Research (April - May):

- Focus: Model optimization and comparative performance testing.
- Tasks: Tuning and training the TransNeXt model to test the project hypothesis (Apr-May).
- Milestone: Poster & Abstracts submission on May 31, 2026.

Phase 4: Finalization & Submission (June - July):

- Focus: Result validation, final documentation, and presentation.
- Tasks: Comprehensive Results Analysis and writing the Final Report (Jun-Jul).
- Milestones: * Final Presentation: June 21, 2026.
    - Final Submission: July 26, 2026.

10.4.2.13 References

[1] S. Sharma and K. Guleria, "Deep Learning Models for Image Classification: Comparison and Applications," in 2022 2nd International Conference on Advance Computing and Innovative Technologies in Engineering (ICACITE), 2022.

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[4] T. Ridnik et al., "TResNet: High Performance GPU-Dedicated Architecture," arXiv:2003.13630, 2020.

[5] M. Tan and Q. V. Le, "EfficientNetV2: Smaller Models and Faster Training," arXiv:2104.00298, 2021.

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